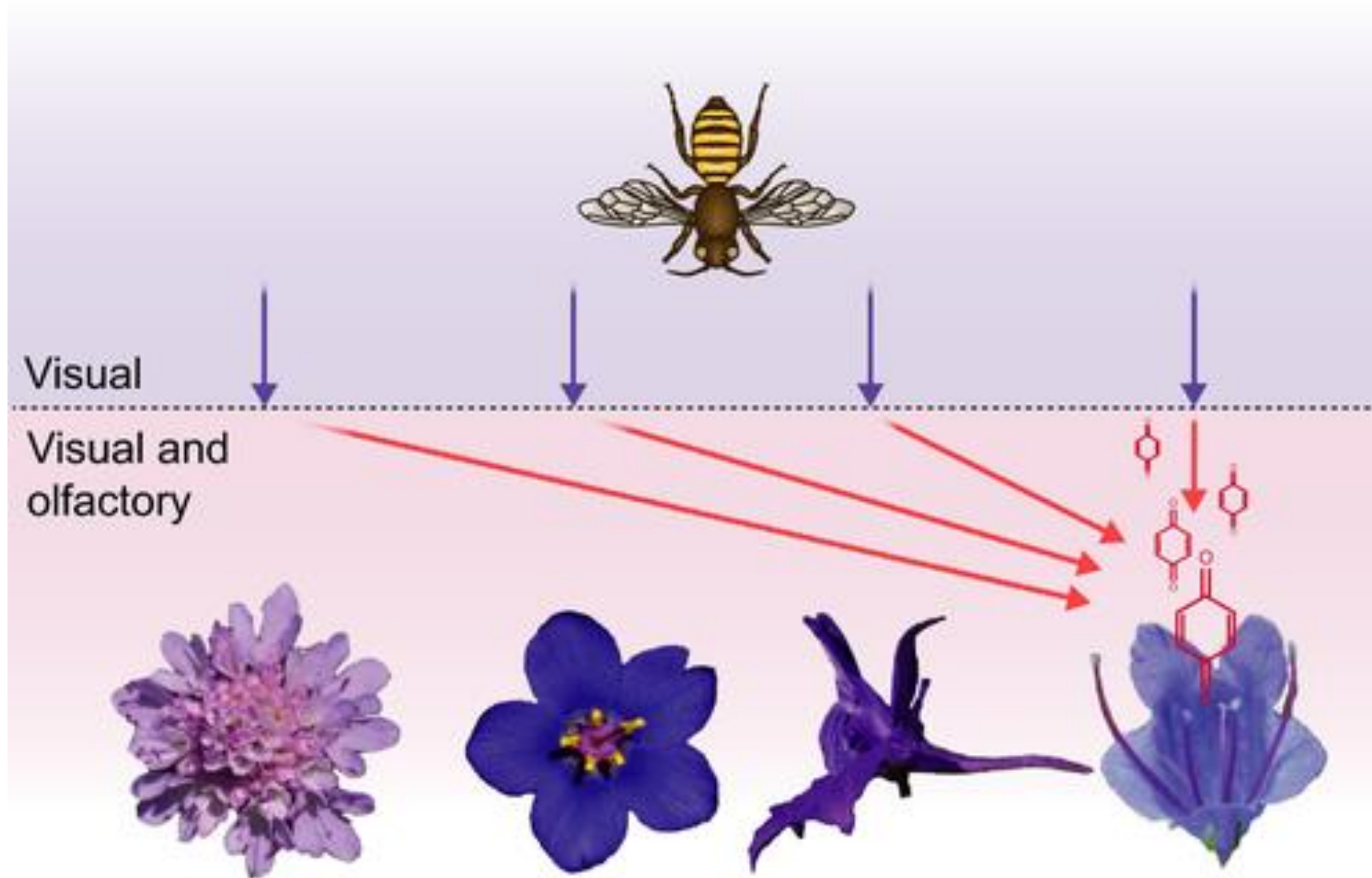


Floral scents

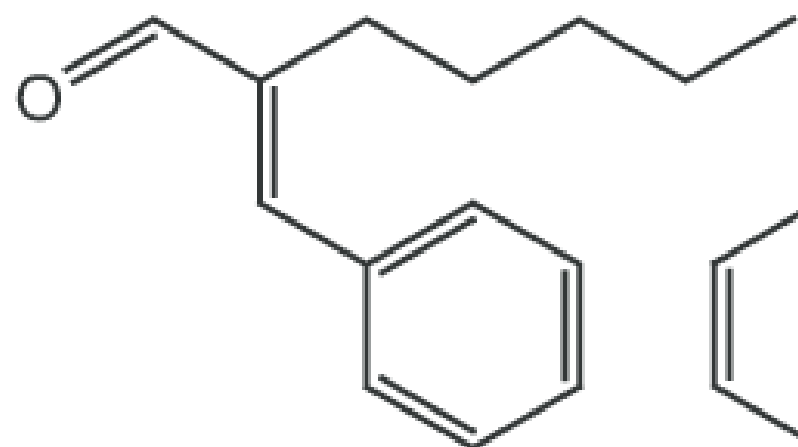


The odour or scent of a flower often plays a major role as attractant to pollinating insects in the angiosperms. Because of the sensitivity of insects to small concentrations of volatile chemicals, flower odours are probably effective at relatively low concentrations. Many species which do not appear to be strongly scented to human senses may, in fact, produce sufficient odour to attract bees or butterflies. In many species, maximum scent production is co-ordinated with the time when the pollen is ripe and the flower is ready for pollination. Diurnal variations in production also occur so that scent is produced for day-time pollinators at noon, for night-time pollinators at dusk.

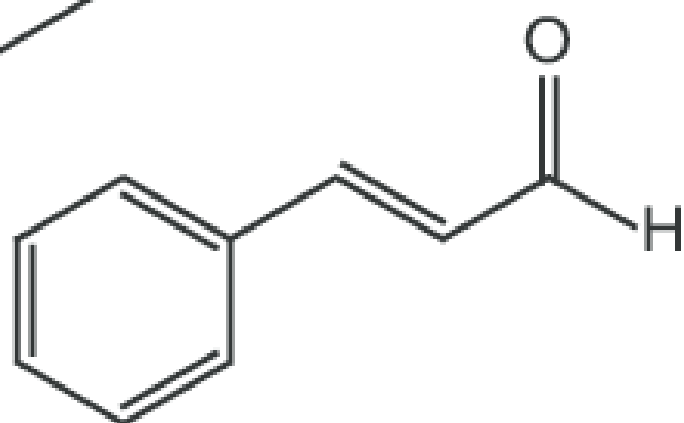




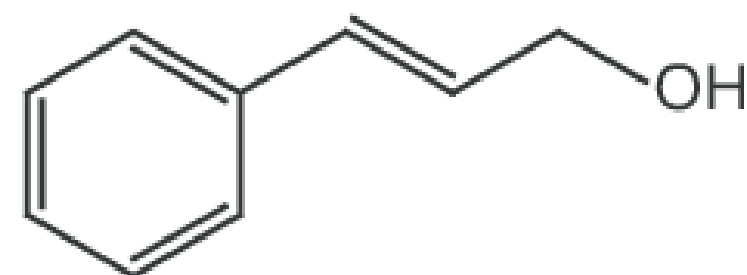
Many pollinators are specialized on a given (group of) plant(s) and are thus dependent on signals or cues to efficiently identify them. Floral color may rarely be sufficiently specific to serve as an identifying token for a specialized visitor



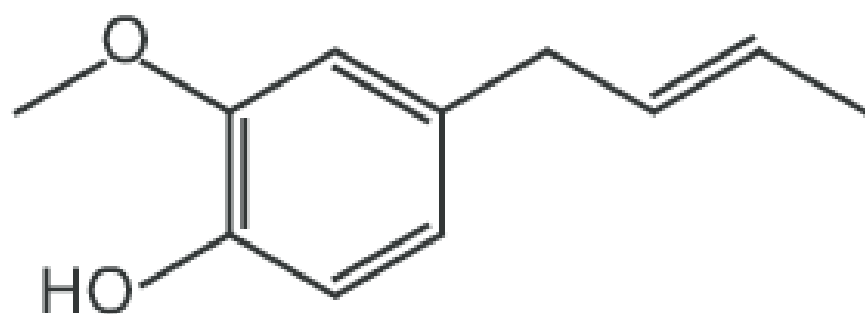
(α)-amylcinnamaldehyde



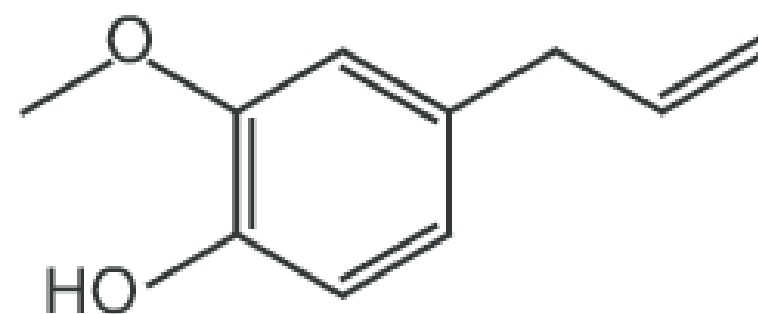
cinnamaldehyde



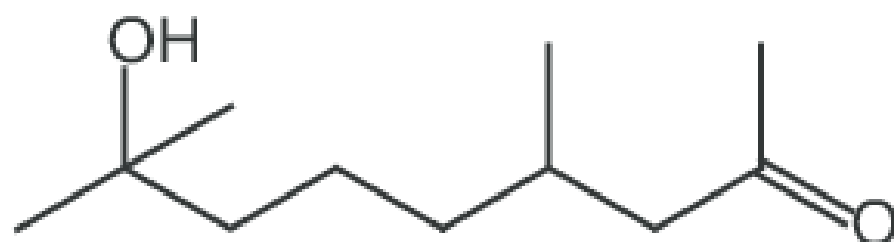
cinnamyl alcohol



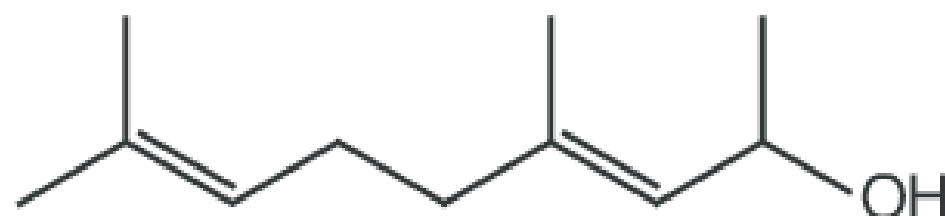
isoeugenol



eugenol



7-hydroxycitronellal



géranol

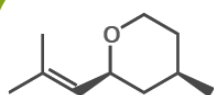
AROMA COMPOUNDS IN COMMON FLOWERS

A wide range of compounds contribute to the scents of flowers. This graphic looks at a selection of major contributors for a number of common flowers. Note that volatile aroma compounds can vary significantly between species; this graphic represents a broad overview of common components, and is by no means definitive!

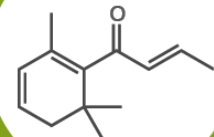


Roses

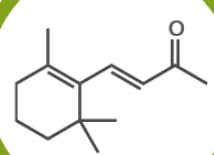
(E)-CIS-ROSE OXIDE



B-DAMASCENONE

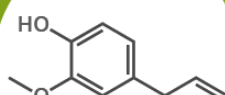


B-IONONE

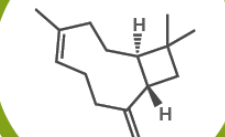


Carnations

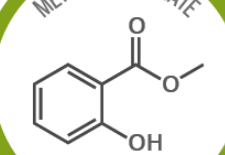
EUGENOL



B-CARYOPHYLLENE

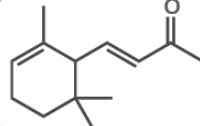


METHYL SALICYLATE

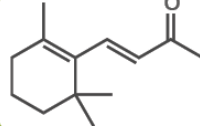


Violets

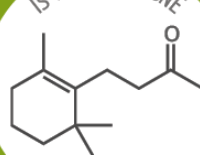
α-IONONE



B-IONONE

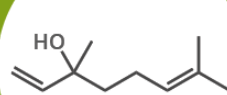


B-DIHYDROIONONE

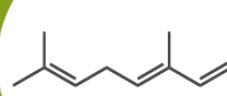


Lilies

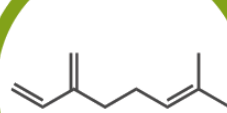
LINALOOL



(E)-B-OCIMENE

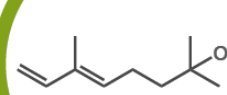


MYRCENE

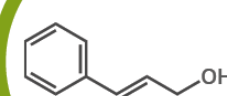


Hyacinth

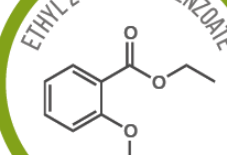
OCIMENOL



CINNAMYL ALCOHOL



ETHYL 2-METHOXYBENZOATE

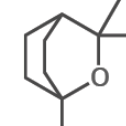


Chrysanthemums

α-PINENE



EUCALYPTOL

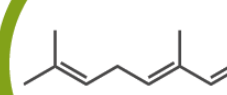


CHRYSANTHENONE

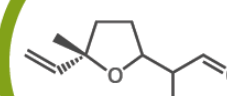


Lilacs

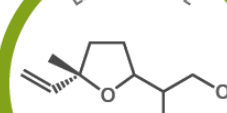
(E)-B-OCIMENE



LILAC ALDEHYDE



LILAC ALCOHOL



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Table 1 Diversity of roles played by floral scent in pollination biology

Mode of action	Example
Direct effects	
Distance attractant	<i>Helicoverpa</i> moths fly upwind to floral odors
Landing cue	<i>Bombus</i> land in response to <i>Rosa</i> pollen odors
Feeding cue (innate)	Naive butterflies probe in response to floral odors
Feeding cue (conditioned)	Scent-conditioned honey bees extend proboscis
Nectar guide	<i>Osmia</i> bees detect nectar remotely using its scent
Repellent	<i>Pieris</i> butterflies repelled by γ -decalactone in <i>Osmanthus</i>
Indirect effects	
Synergist with visual/gustatory cues	Synergizes <i>M. sexta</i> visual feeding response
Attention enhancement	Sharpens visual discrimination by <i>B. terrestris</i>
Host cue for florivores	Attracts flower-feeding insects to <i>Cirsium arvense</i>
Attractant for predators	Attracts crab spiders to flowers favored by honey bees
Repellent for predators	<i>Acacia</i> pollen odor repels ant guards during bee visits
Artifact of wounds or induced defense	<i>Pastinaca</i> fragrance changes after florivory
Artifact of microbial infestation	in situ nectar fermentation in <i>Agave palmeri</i> flowers

<https://app.jove.com/v/2282/behavioural-pharmacology-classical-conditioning-proboscis-extension>

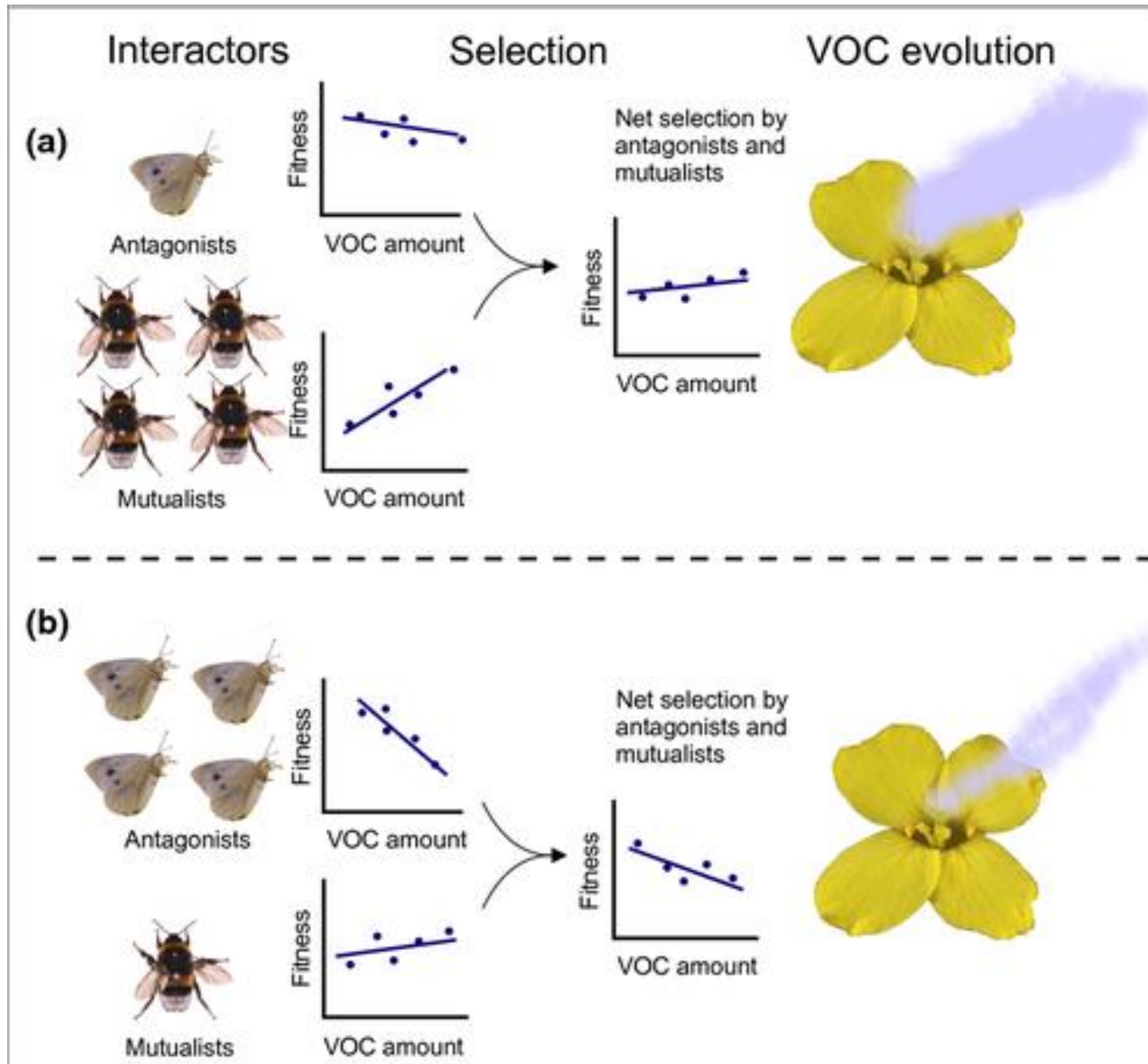
Floral scent may rarely be detectable by mutualistic visitors only, but often also attracts antagonists, such as herbivores or florivores. Therefore, it can be assumed that VOCs evolve under conflicting selection for reproduction and survival, much like sexual signals in animals.



1,4-dimethoxybenzene, s
the major floral volatile and
attracts specialized
squash bees as
pollinators, as well as
striped cucumber beetles,
a major herbivore and
florivore of this plant



This study is the first to demonstrate that enhanced floral fragrance can increase the attraction of detrimental florivores and decrease plant reproduction, suggesting that florivory as well as pollination has shaped the evolution of floral scent.



Depending on the abundance of pollinators and herbivores, the net fitness outcome of volatile emission differs, leading to a spatially varying selection mosaic on floral scent. In (a), herbivores are rare, and thus the net selection on the volatile organic compound (VOC) is positive and strong scent emission evolves. In (b), herbivores are abundant, leading to net negative selection and the evolution of weakly scented flowers.

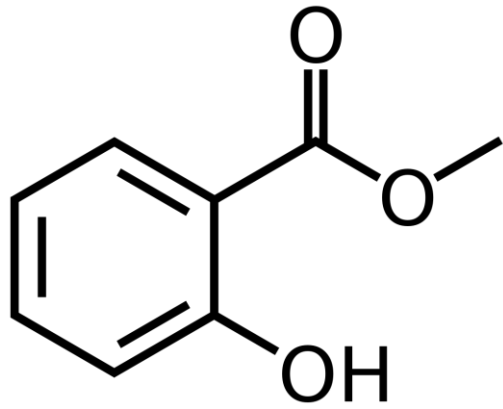
Typically, animal pollinators are attracted to flowers by sensory stimuli in the form of pigments, volatiles, and cuticular substances (hairs, waxes) derived from plant secondary metabolism. Sometime, primary plant metabolites, such as respiratory carbon dioxide (CO_2), may also function as pollinator attractants.

Datura wrightii, whose flowers open at sunset and emit high CO_2 during the first 30–60 min, coinciding with peak nectar availability

Manduca sexta moths utilize floral CO_2 as a redundant distance attractant in wind-tunnel experiments, but from close range, males prefer scented flowers with above-ambient CO_2 , whereas females do so only in the presence of host-plant odors. This result suggests that *M. sexta* females utilize floral CO_2 as well as nectar rewards to evaluate host-plant quality while drinking nectar from *Datura* or *Nicotiana* plants.

Such floral scents are often called “Context specific cue”

Context dependent information



methylsalicylate



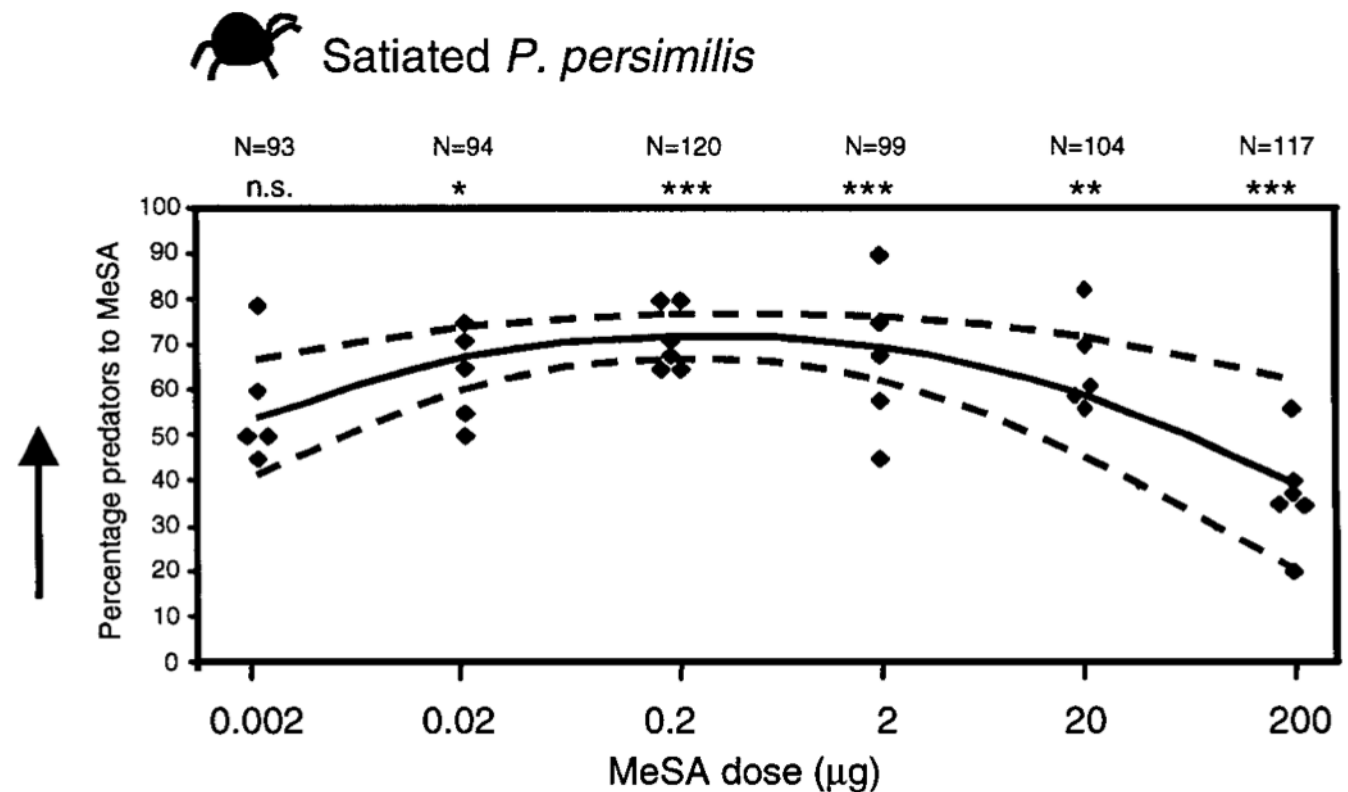
Andersson et al 2000, Proc. Roy. Soc. B

Pieris napi males transfer methyl-salicylate (MeSA), while mating. The mated females emit when courted and which makes other males quickly abandon them. MeSA acts as an anti-aphrodisiac.



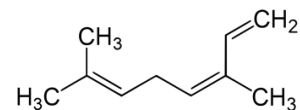
DeBoer et al J. Chem. Ecol. 2004

Methyl salicylate elicits hunting behaviour in carnivorous mites.

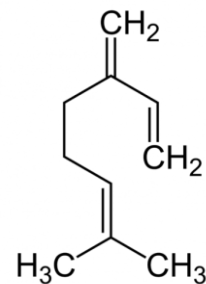




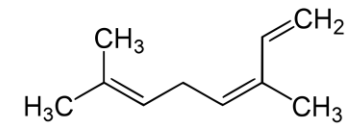
Macrozamia lucida



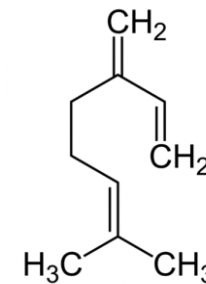
Ocimene



Myrcene



Ocimene



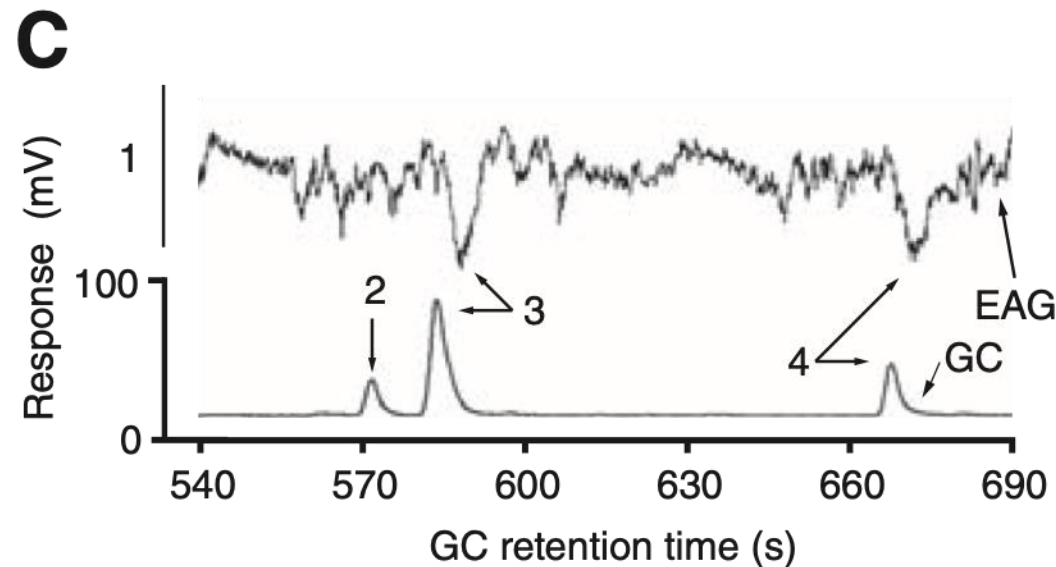
Myrcene



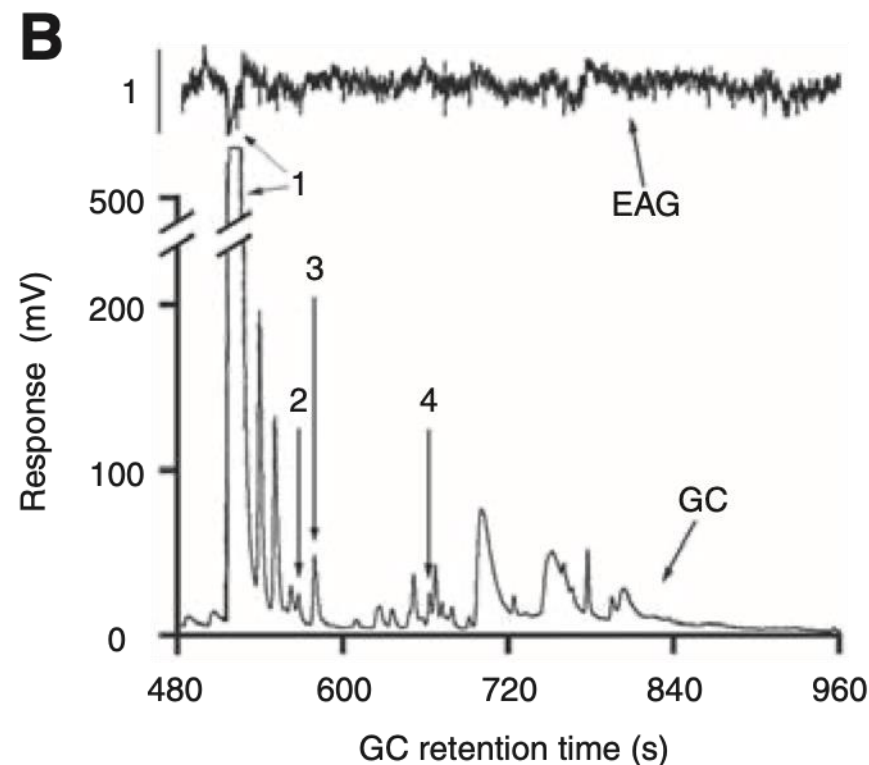
Cycadothrips

Australian cycad *Macrozamia lucida* is pollinated by *Cycadothrips*, which are attracted to low levels of β -myrcene and (*E*)- β -ocimene. However, the cones are thermogenic, and they synchronize mid-day heat production with a million-fold increase in scent emissions, forcing thrips to depart the cones. They are then attracted to the female megasporophylls that emit the same compounds in low concentrations. This obligate pollination system is mediated through dosage-dependent pollinator attraction and repellence by same volatile compounds.

Odor-Mediated Push-Pull Pollination in Cycads



This is characterised as a “push-pull” pollination strategy. Flowers are generally portrayed as only “pulling” pollinators via visual or odor cues. Here, chemicals are repelling the pollinators after pollination.



Floral scent may have originally evolved to deter herbivores, and this system may represent a conserved early intermediary in the evolution of seed plant pollination.

Terry et al Science, 2007

Sensory exploitation (sensory drive, pre-existing bias) suggests that pollinators select for VOCs that are primarily used in ecological contexts other than flower visitation.

Methoxylated aromatic compounds have been identified as components of sex or aggregation pheromones in scarab beetles but also as attractive floral volatiles in cyclocephaline scarab-pollinated Araceae. Such co-option of chemical signals can be explained by sensory drive, namely through pollinator-mediated selection for volatiles for which olfactory receptors and/or preferences have not evolved in the pollination context. Because the pheromone functions of methoxylated aromatics are evolutionarily older than their occurrence as floral volatiles, this is an example of pre-existing bias

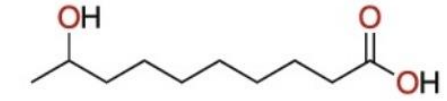
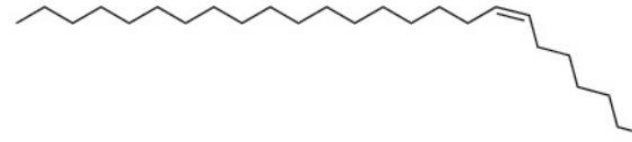
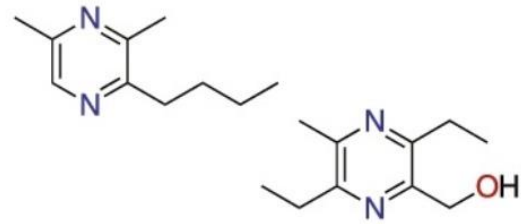
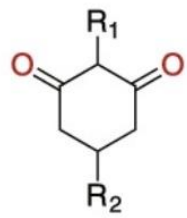


Herbivore-induced GLVs are used by social wasps when hunting for caterpillars on plants, and have seemingly been co-opted to recruit these insects as pollinators through sensory exploitation. Although the flowers of both *S. umbrosa* and *E. helleborine* reward their visitors with nectar, those of *Epipactis veratrifolia* produce no reward, but emit terpenoids that constitute the alarm pheromone of aphids. This signal attracts hover flies that lay eggs on flowers (their larval stage feeds on aphids) and also pollinate the flowers. Thus, pre-existing bias can, but does not always, lead to the evolution of rewardless flowers.

Upon herbivore damage, plants release a specific blend of volatile organic compounds that are attractive to predators of the herbivore. These are called herbivore Induced Plant Volatiles (HIPV).



Pollination by sexual deception - It takes chemistry to work



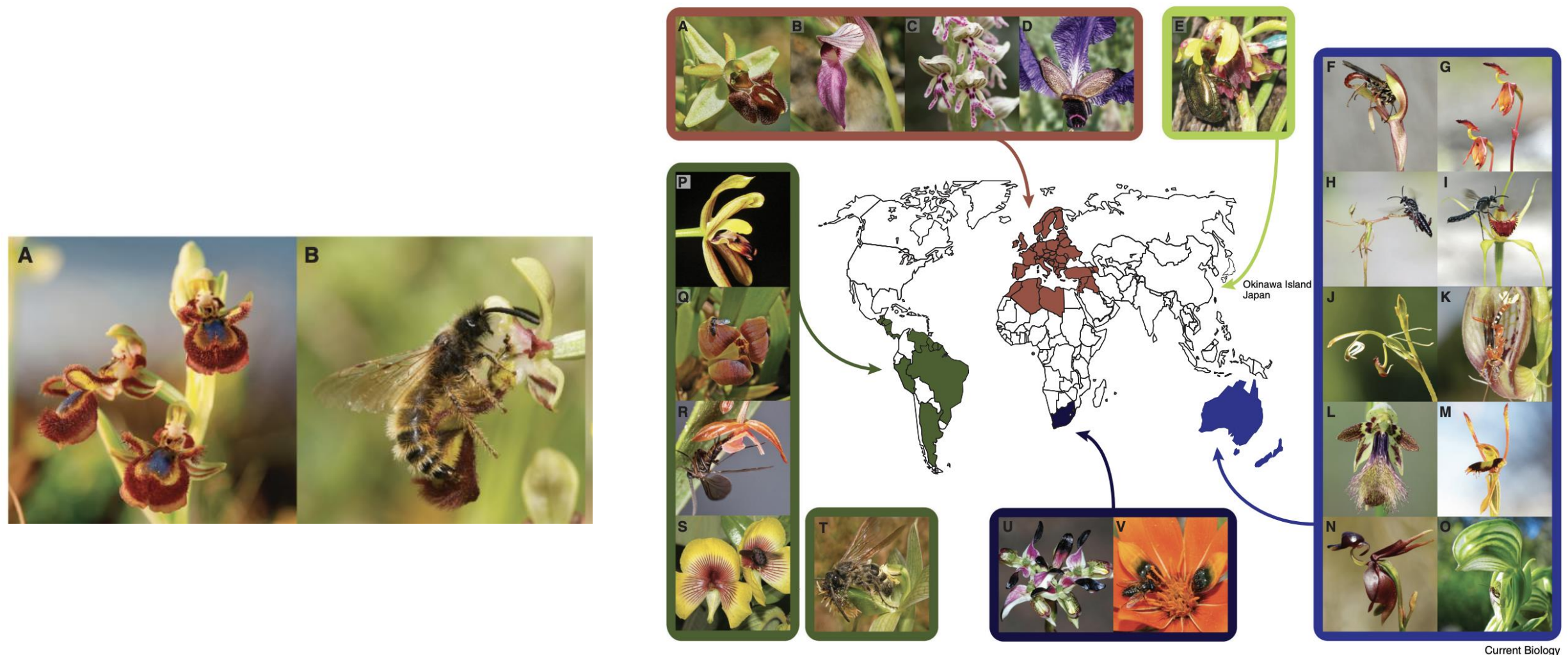
87% of plants depend on animals for pollination, with most of these paying for the service of pollination via food rewards of nectar or pollen. As in human economic systems, however, some cheating and deception occurs, with the pollination strategy of sexual deception being one such example.

Deceptive pollination



Sexually deceptive plants secure pollination via sexual mimicry, which offers the false promise of a mating opportunity to potential pollinators.

In Algeria, Maurice-Alexandre Pouyanne observed the pollination of the insect-like orchid, *Ophrys speculum*, over many years: his observations led him to the astonishing conclusion that the male scoliid wasp pollinators were so fooled by the orchid that they engaged in attempted copulation with the flower

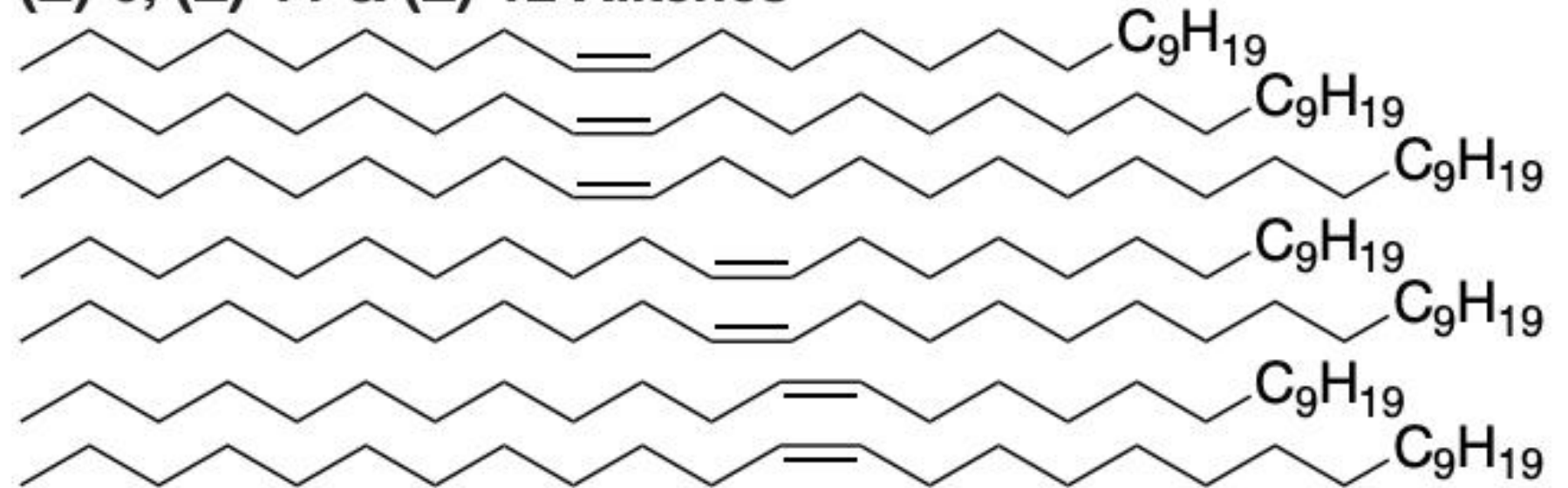


The first experimental confirmation of sex pheromone mimicry was published in 1999, for the study species *Ophrys sphegodes*

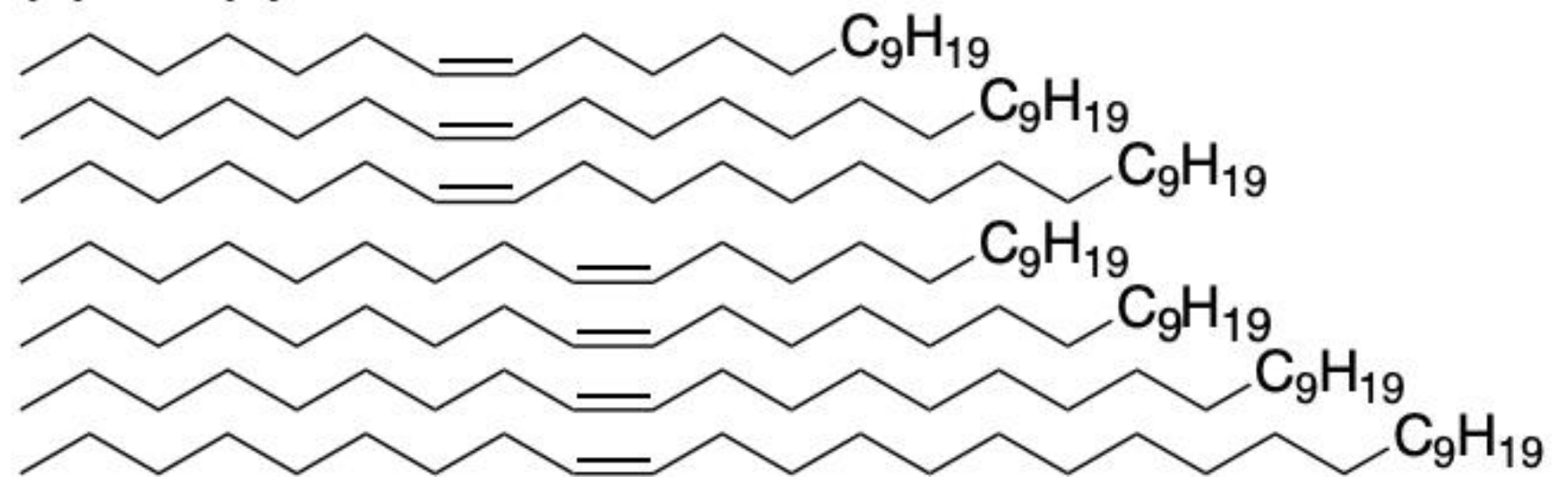
Solitary Bees



(Z)-9, (Z)-11 & (Z)-12 Alkenes



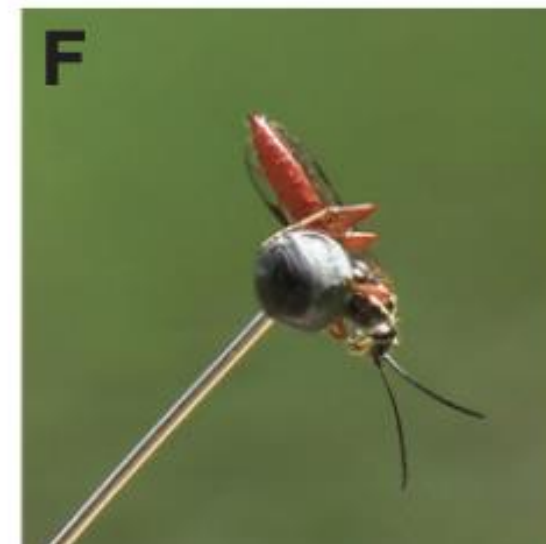
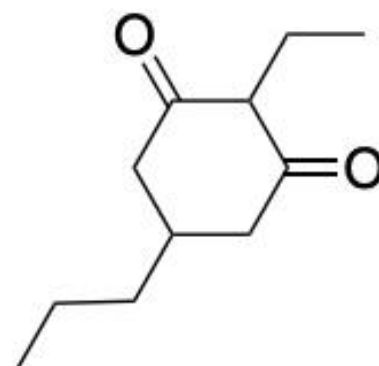
(Z)-7 & (Z)-9 Alkenes



Thynnine Wasps



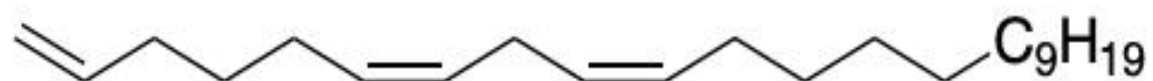
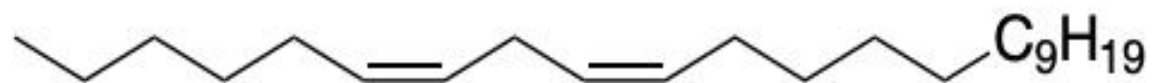
Cyclohexanedione (chiloglottone 1)



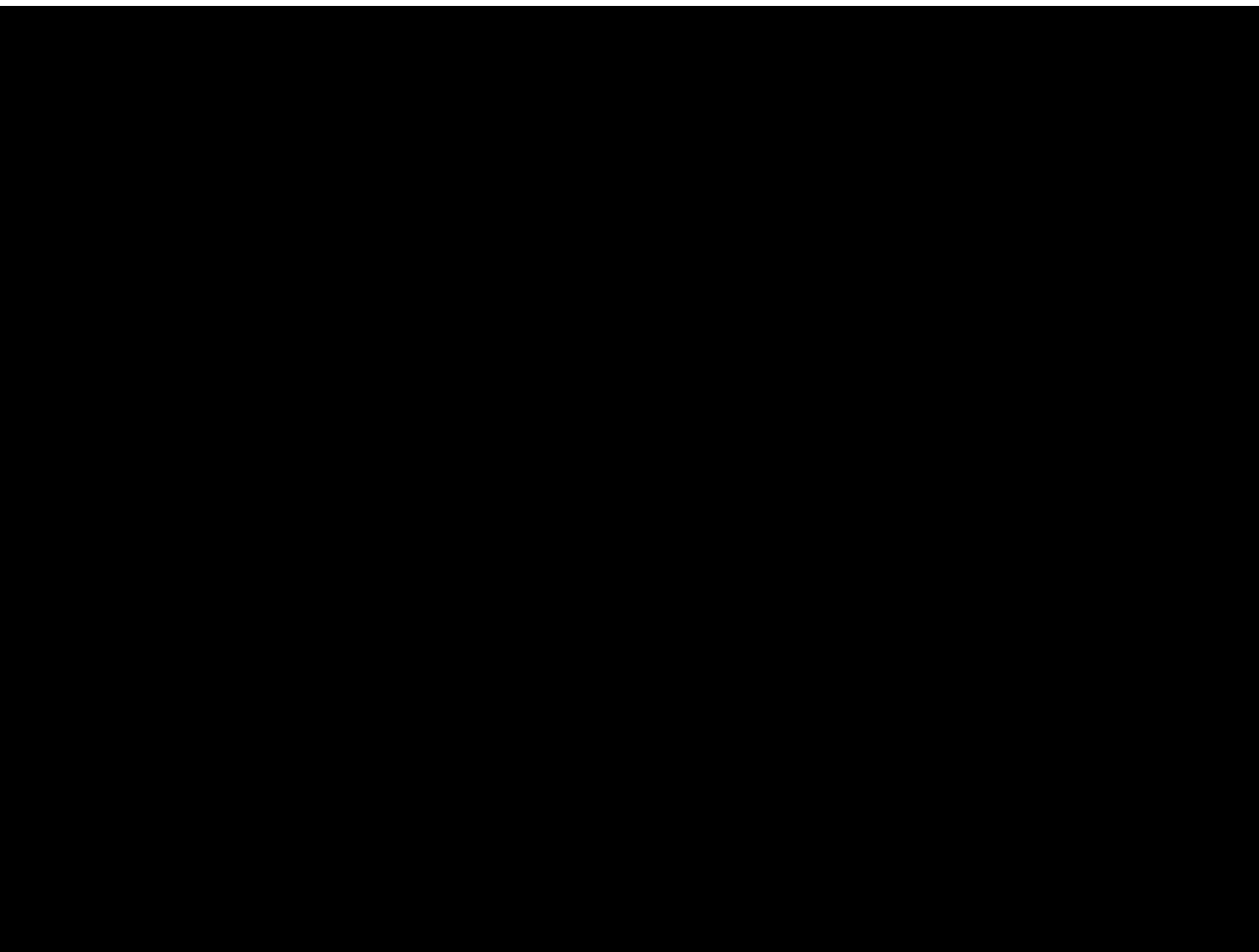
Fungus Gnat



(6Z, 9Z)-Diene and (6Z, 9Z)-Triene

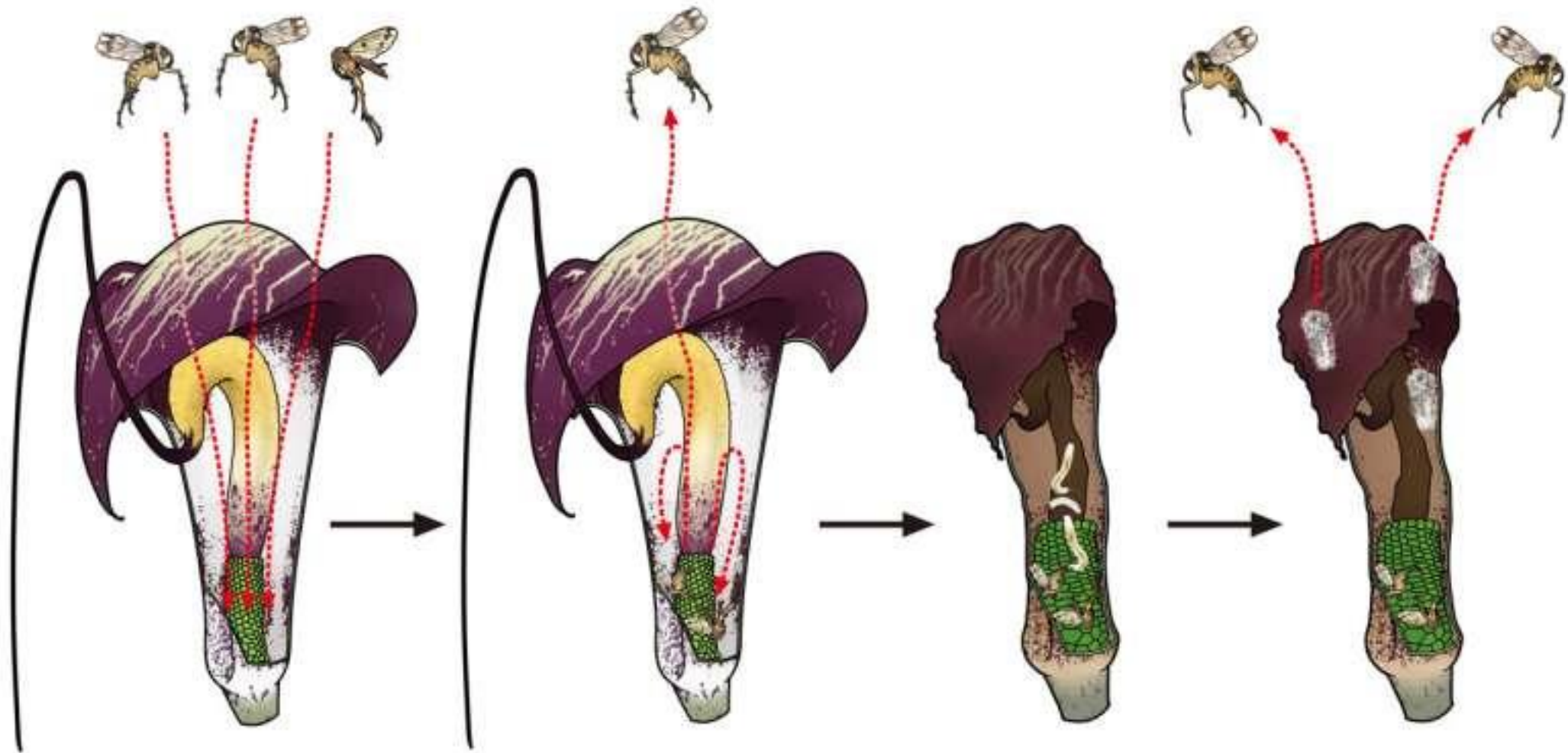


In the gnat pollinated *Pterostylis orbiculata*, an unusual long-chain diene and triene are the key to sexual mimicry



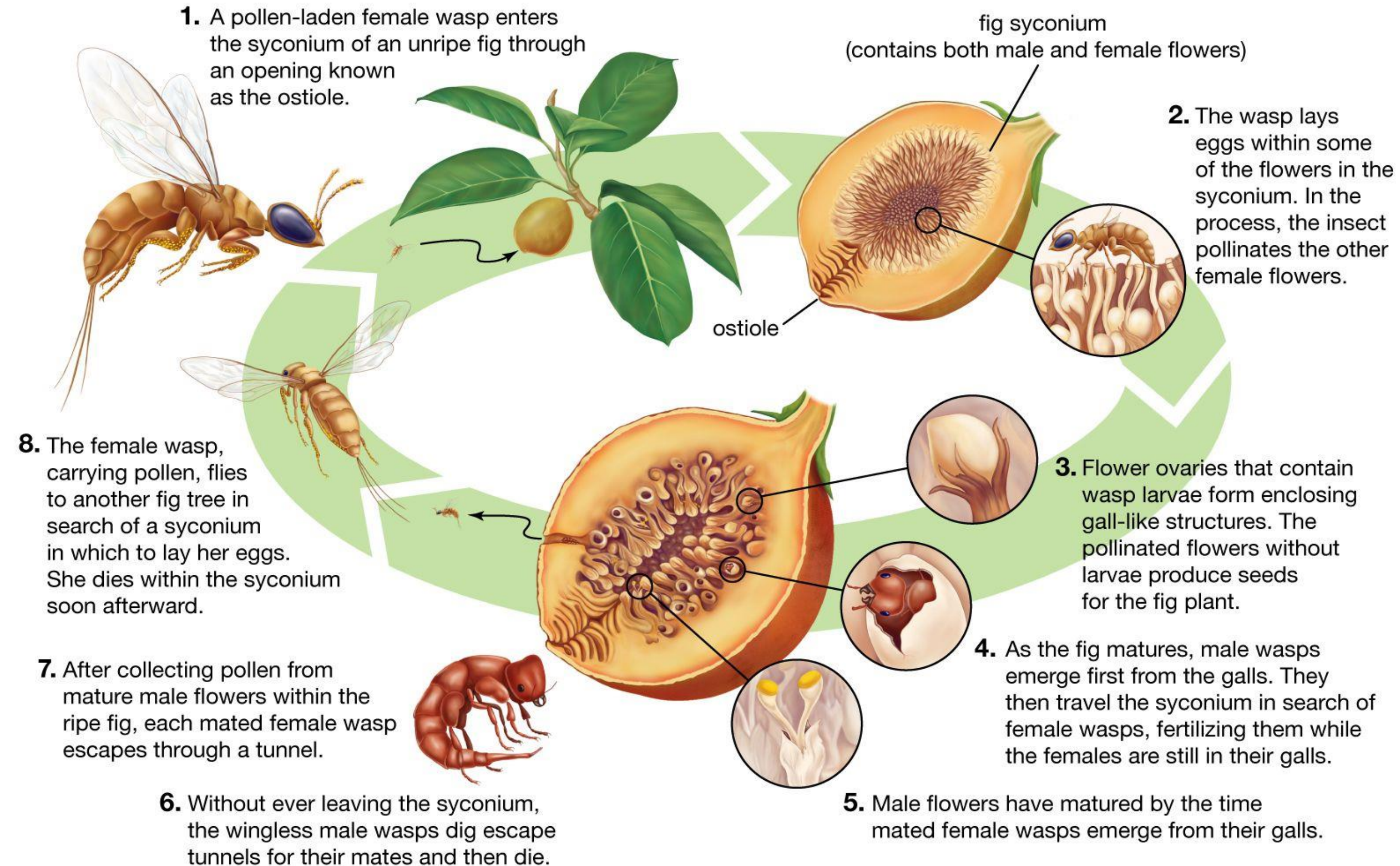


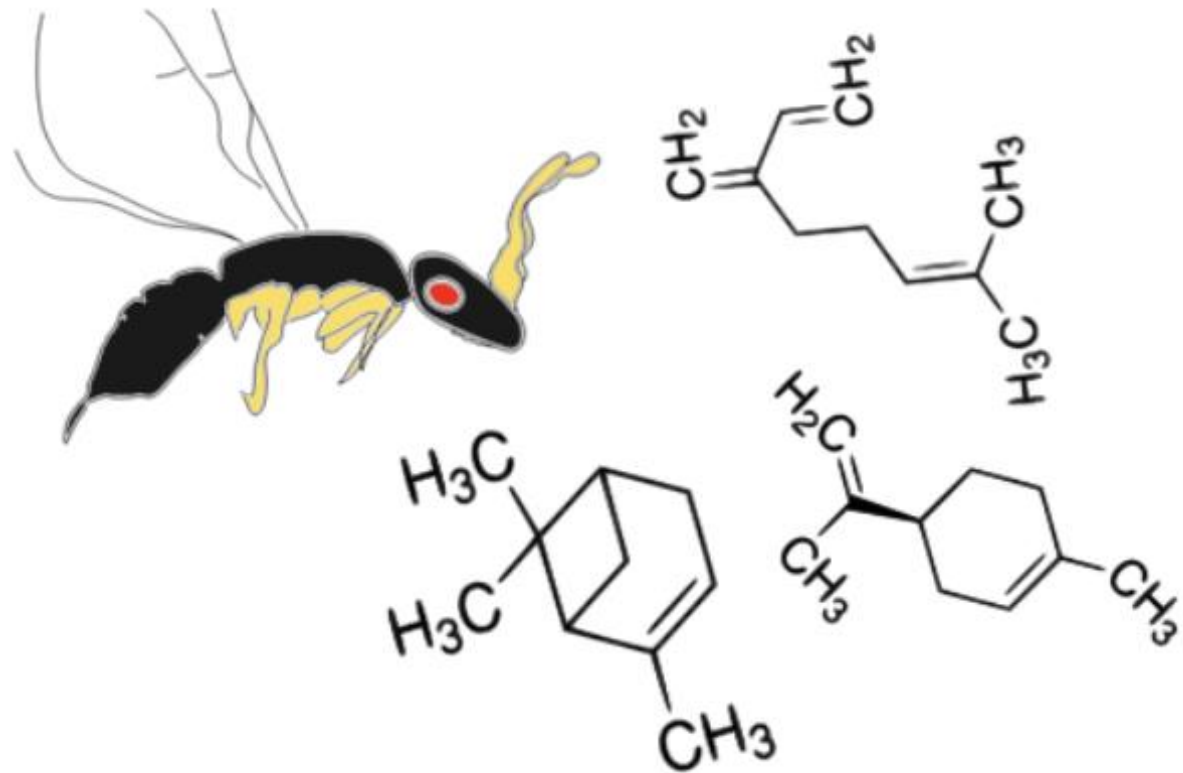
Nursery pollination involves pollinators that lay eggs on the flowers they pollinate and have their brood fed on flower parts or developing ovules



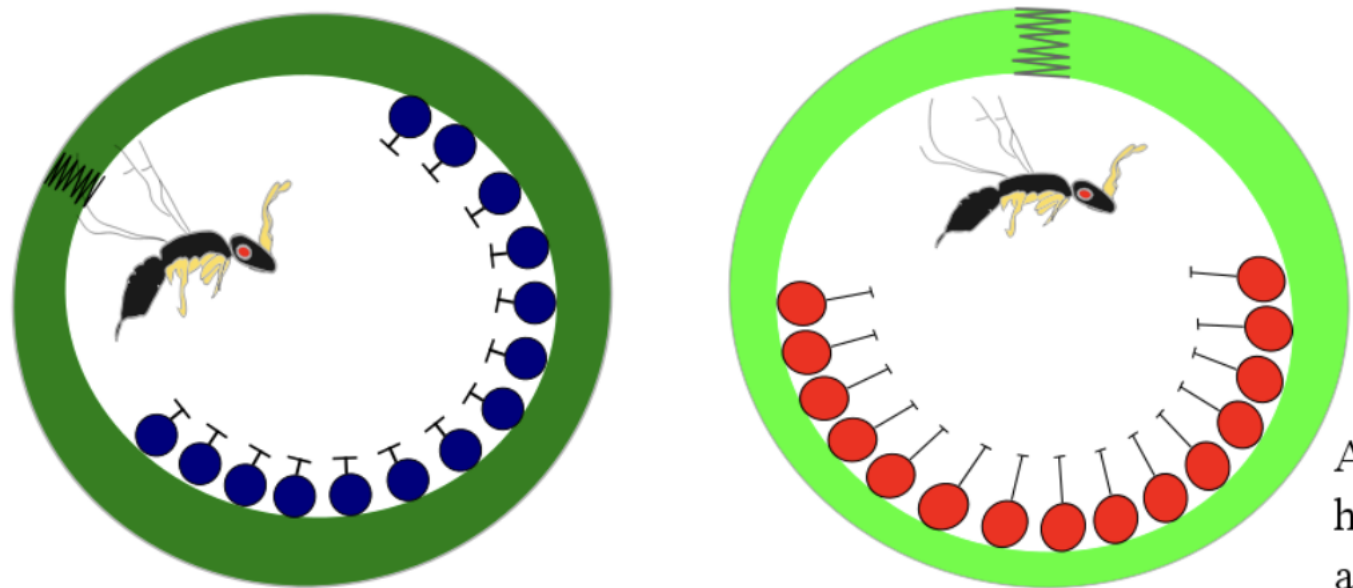
The plant uses a musky odor to lure fungus gnats that usually feed and lay their eggs on mushrooms into their cup-shaped flowers. The insects can escape from the male flowers, but only after tribulations that will result in them being covered in pollen. However, there is no escape from the female flowers. Once they enter these, the involuntary pollen carriers struggle to find an exit, which ensures that they will pollinate the flower, but they can't get a hold on the waxy interior and perish.

In nursery pollination mutualisms, which are usually obligate interactions, olfactory attraction of pollinators by floral volatile organic compounds (VOCs) is the main step in guaranteeing partner encounter.





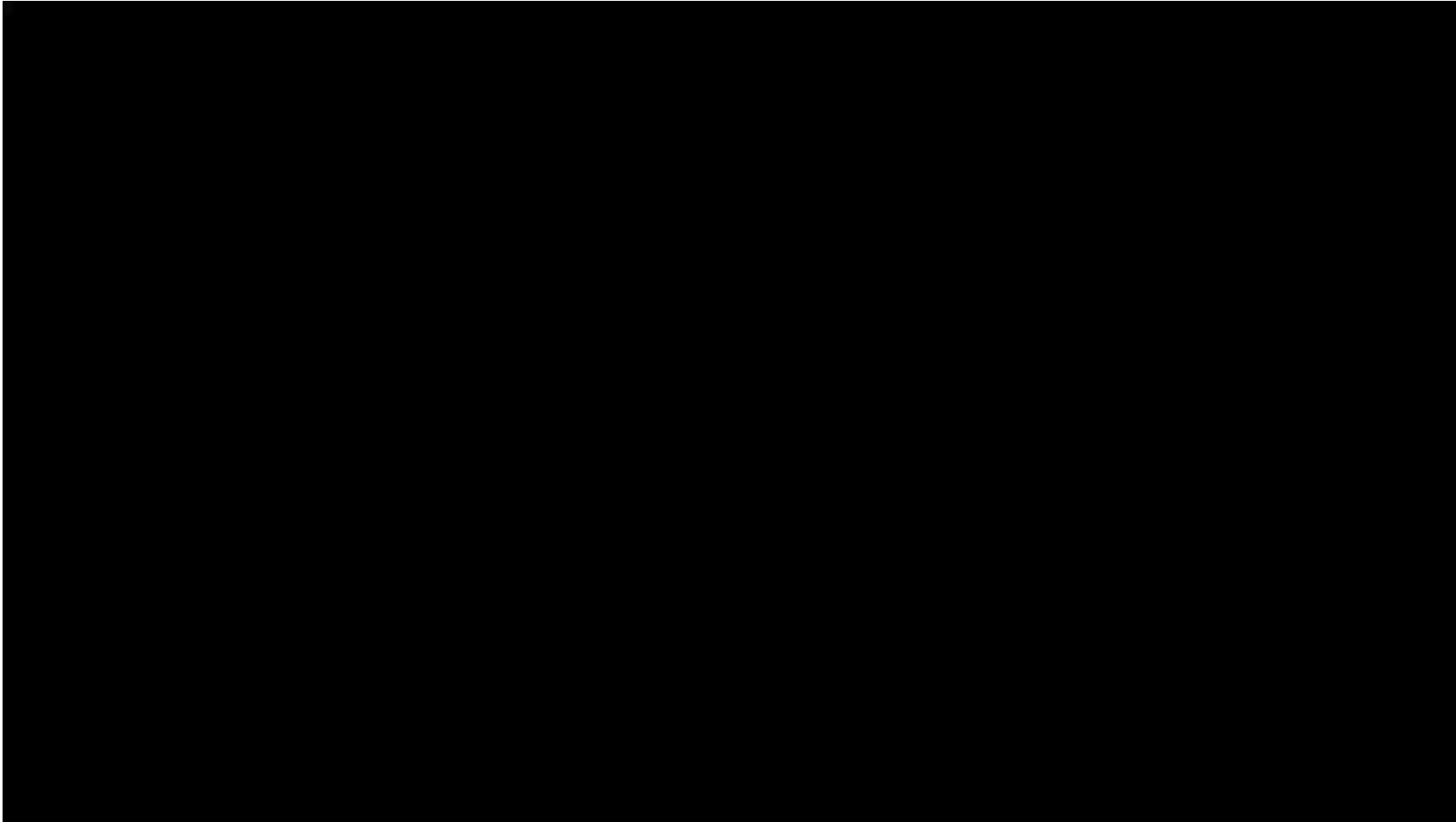
With over 750 described species, the maintenance of species specificity between figs and fig-wasps is believed to depend primarily on unique volatile signals which attract fig-wasps to their host fig species. Even closely related fig species have significantly different scent profiles, and indeed even subspecies have different VOC compositions from each other.



A
h
a

Additionally style length has to be compatible for successful oviposition. Alternative fig species has equally attractive volatiles, however, style-ovipositor length incompatibility prevent the atypical wasp from laying their eggs – representing a significant cost for the wasp fitness.



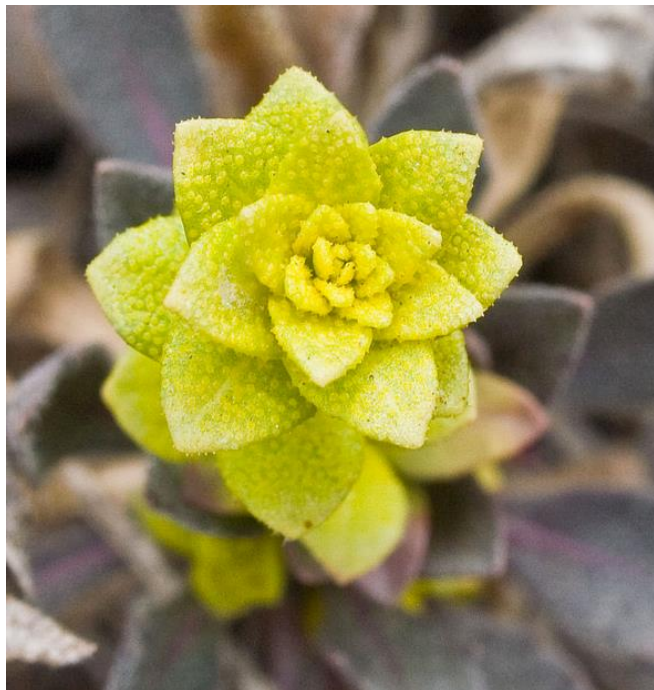


In nursery pollination mutualisms, pollinators reproduce within the inflorescence they pollinate. In these systems, each partner depends directly on the other for its reproduction. Therefore, the signal responsible for partner encounter is crucial in these horizontally transmitted mutualisms, in which the association between specific partners must be renewed at each generation.

The orchid *Gastrodia similis* has flowers that smell similar to fermented apple or pineapple, and are pollinated by the fruit fly *Scaptodrosophila bangi*. The floral scent contains **ethyl acetate**, **ethyl isobutyrate** and **methyl isobutyrate**, which were shown to attract only the pollinating fly species (Martos *et al.*, 2015).



These volatile compounds were also identified in the odor of fermenting indigenous figs found in the surrounding understory, and these figs were highly attractive brood sites of *S. bangi* (Martos *et al.*, [2015](#)). Furthermore, the size, shape, microtexture and dull-brown coloration of the flowers likely contribute to mimicry of the figs (Martos *et al.*, [2015](#)).



Chemical class compounds	<i>P. monoica</i> on <i>A. holboellii</i>	<i>P. consimilus</i> on <i>A. crandallii</i>	<i>P. consimilus</i> on <i>A. demissa</i>	<i>P. thlaspeos</i> on <i>A. gunnisoniana</i>	<i>P. thlaspeos</i> on <i>A. holboellii</i>
Aromatics					
Benzaldehyde	3.4%*	2.1	1.2	1.3	<1.0
Phenylacetaldehyde	61.6	47.5	30.6	35.9	<1.0
2-phenylethanol	11.9	3.2	2.7	24.6	13.2
Methyl benzoate	1.5	<1.0	2.0	1.7	
Benzyl acetate		60.5			
Phenethyl acetate	11.3	29.0		16.4	<1.0
Phenethyl propionate	<1.0	1.0	<1.0	1.1	
Phenethyl isobutyrate	<1.0	<1.0	<1.0	<1.0	
Phenethyl isovalerate†	5.2	12.7	1.2	9.6	<1.0
Phenethyl senecioate†	<1.0	<1.0	<1.0	1.8	
Amyl salicylate	1.0	<1.0	<1.0	<1.0	

The authors have shown that *Puccinia* pseudoflowers use fragrance to attract bees and flies that provide an essential ‘pollination’ service during the sexual cycle of these fungi. Our surveys of scent production by *Puccinia* pseudoflowers, their *Arabis* hosts and coblooming true flowers demonstrate that pseudoflower fragrance is chemically distinct and consists entirely of aromatic alcohols, aldehydes and esters.

Raguso & Roy, 1998, Mol. Ecol.

